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1 Working with Matrices and Vectors

1.1 Notation for Scalars, Vectors, and Matrices

Lowercase letters $\Rightarrow$ scalars: x ; c ; σ .

Boldface, lowercase letters $\Rightarrow$ vectors: $\mathbf{x}$; $\mathbf{y}$; $\boldsymbol{\beta}$.

Boldface, uppercase letters $\Rightarrow$ matrices: $\mathbf{A}$; $\mathbf{X}$; $\boldsymbol{\Sigma}$.

1.2 Matrix and Vector Operations

Definition 1.

A column of real numbers is called a **vector**.

Example 1.

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} \quad \mathbf{a} = \begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix} \quad \mathbf{1} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

Since $\mathbf{y}$ has n elements it is said to have **order** (or dimension) n .

Definition 2.

A rectangular array of elements with m rows and k columns is called an $m \times k$ **matrix**.

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1k} \\ a_{21} & a_{22} & \cdots & a_{2k} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mk} \end{bmatrix}$$

This matrix is said to be of **order** (or dimension) $m \times k$, where

- m is the **row** order (dimension)
- k is the **column** order (dimension)

Example 2.

$$\mathbf{A} = \begin{bmatrix} 1 & 3 & -2 \\ 0 & 4 & 5 \end{bmatrix} \quad \mathbf{I} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix}$$

R-code:

```
A = matrix(c(1,3,-2,0,4,5), 2,3, byrow = T)
A
I = diag(3)
I
B = matrix(c(1,3,2,6), 2,2, byrow = T)
B
```

Definition 3. Matrix addition

If $\mathbf{A}$ and $\mathbf{B}$ are both $m \times k$ matrices, then

$$\mathbf{C} = \mathbf{A} + \mathbf{B}$$

$$= \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1k} \\ a_{21} & a_{22} & \cdots & a_{2k} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mk} \end{bmatrix} + \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1k} \\ b_{21} & b_{22} & \cdots & b_{2k} \\ \vdots & \vdots & & \vdots \\ b_{m1} & b_{m2} & \cdots & b_{mk} \end{bmatrix}$$

$$= \begin{bmatrix} a_{11} + b_{11} & a_{12} + b_{12} & \cdots & a_{1k} + b_{1k} \\ a_{21} + b_{21} & a_{22} + b_{22} & \cdots & a_{2k} + b_{2k} \\ \vdots & \vdots & & \vdots \\ a_{m1} + b_{m1} & a_{m2} + b_{m2} & \cdots & a_{mk} + b_{mk} \end{bmatrix}$$

Notation:

$$C_{m \times k} = \{c_{ij}\} \text{ where } c_{ij} = a_{ij} + b_{ij}$$

Definition 4. Matrix subtraction

If $\mathbf{A}$ and $\mathbf{B}$ are $m \times k$ matrices, then $\mathbf{C} = \mathbf{A} - \mathbf{B}$ is defined by

$$\mathbf{C} = \{c_{ij}\} \text{ where } c_{ij} = a_{ij} - b_{ij}.$$

Example 3.

$$\begin{bmatrix} 3 & 6 \\ 2 & 1 \end{bmatrix} + \begin{bmatrix} 7 & -4 \\ -3 & 2 \end{bmatrix} = \begin{bmatrix} 10 & 2 \\ -1 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -1 \\ 1 & 1 \\ 1 & 0 \end{bmatrix} - \begin{bmatrix} 1 & -1 \\ 2 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ -1 & 1 \\ 0 & -1 \end{bmatrix}$$

R-codes:

```
A = matrix(c(3,6,2,1), 2,2,byrow=T)
B = matrix(c(7,-4,-3,2), 2,2,byrow=T)
C = A+B
D = matrix(c(1,-1,1,1,1,0), 3,2,byrow=T)
E = matrix(c(1,-1,2,0,1,1), 3,2,byrow=T)
F = D-E
```

Definition 5. Scalar multiplication

Let a be a scalar and $\mathbf{B} = \{b_{ij}\}$ be an $m \times k$ matrix, then

$$a\mathbf{B} = \mathbf{B}a = \{ab_{ij}\}$$

Example 4.

$$2 \begin{bmatrix} 2 & -1 & 3 \\ 0 & 4 & -2 \end{bmatrix} = \begin{bmatrix} 4 & -2 & 6 \\ 0 & 8 & -4 \end{bmatrix}$$

```
R-Code:
A = matrix(c(2,-1,3,0,4,2), 2,3,byrow=T)
B= 2*A
```

Definition 6. Transpose

The transpose of the $m \times k$ matrix $\mathbf{A} = \{a_{ij}\}$ is the $k \times m$ matrix with elements $\{a_{ji}\}$. The transpose of $\mathbf{A}$ is denoted by $\mathbf{A}^T$ (or $\mathbf{A}'$).

Example 5.

$$\mathbf{A} = \begin{bmatrix} 1 & 4 \\ 3 & 0 \\ -2 & 6 \end{bmatrix} \quad \mathbf{A}^T = \begin{bmatrix} 1 & 3 & -2 \\ 4 & 0 & 6 \end{bmatrix}$$

```
R-code:
A = matrix(c(1,4,3,0,-2,6), 3,2,byrow=T)
AT = t(A)
AT
```

Definition 7. If a matrix has the same number of rows and columns it is called a **square matrix**.

$$\mathbf{A}_{k \times k} = \begin{bmatrix} a_{11} & \cdots & a_{1k} \\ \vdots & & \vdots \\ a_{k1} & \cdots & a_{kk} \end{bmatrix}$$

is said to have order (or dimension) k .

Definition 8. A square matrix $\mathbf{A} = \{a_{ij}\}$ is **symmetric** if $\mathbf{A} = \mathbf{A}^T$, that is, if $a_{ij} = a_{ji}$ for all (i, j) .

Example 6.

$$\mathbf{A} = \begin{bmatrix} 4 & 1 \\ 1 & 2 \end{bmatrix}$$

$$\mathbf{B} = \begin{bmatrix} 4 & 3 & 2 & 1 \\ 3 & 5 & 0 & -2 \\ 2 & 0 & 3 & -1 \\ 1 & -2 & -1 & 2 \end{bmatrix}$$

Definition 9. Inner product (crossproduct) of two vectors of order n

$$\begin{aligned} \mathbf{a}^T \mathbf{y} &= [a_1, a_2, \cdots a_n] \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} \\ &= [a_1 y_1 + a_2 y_2 + \cdots + a_n y_n] \\ &= \sum_{j=1}^n a_j y_j \end{aligned}$$

Note that $\mathbf{a}^T \mathbf{y} = \mathbf{y}^T \mathbf{a}$

```
R-codes:
a = c(1, 7, -6, 4)
y = c(2, -2, 1, 5)
aTy1 = t(a)%*%y
aTy2 = a*%*%y
aTy3 = crossprod(a,y)
```

Definition 10. Matrix multiplication

The product of an $n \times k$ matrix $\mathbf{A}$ and a $k \times m$ matrix $\mathbf{B}$ is the $n \times m$ matrix $\mathbf{C} = \{c_{ij}\}$ with elements

$$c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{ik}b_{kj}$$

Example 7.

$$\mathbf{A} = \begin{bmatrix} 3 & 0 & -2 \\ 1 & -1 & 4 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} \quad \mathbf{C} = \mathbf{AB} = \begin{bmatrix} 1 & -3 \\ 4 & 11 \end{bmatrix}$$

R-codes:

```
A = matrix(c(3,0,-2,1,-1,4), 2,3,byrow=T)
B = matrix(c(1,1,1,2,1,3), 3,2,byrow=T)
C = A%*%B
```

Definition 11. Elementwise multiplication of two matrices

$$\begin{aligned} \mathbf{A} \# \mathbf{B} &= \begin{bmatrix} a_{11} & \cdots & a_{1m} \\ \vdots & & \vdots \\ a_{k1} & \cdots & a_{km} \end{bmatrix} \# \begin{bmatrix} b_{11} & \cdots & b_{1m} \\ \vdots & & \vdots \\ b_{k1} & \cdots & b_{km} \end{bmatrix} \\ &= \begin{bmatrix} a_{11}b_{11} & \cdots & a_{1m}b_{1m} \\ \vdots & & \vdots \\ a_{k1}b_{k1} & \cdots & a_{km}b_{km} \end{bmatrix} \end{aligned}$$

Example 8.

$$\begin{bmatrix} 3 & 1 \\ 2 & 4 \\ 0 & 6 \end{bmatrix} \# \begin{bmatrix} 1 & -5 \\ -3 & 4 \\ -2 & 2 \end{bmatrix} = \begin{bmatrix} 3 & -5 \\ -6 & 16 \\ 0 & 12 \end{bmatrix}$$

R-codes:

```
A = matrix(c(3,1,2,4,0,6), 3,2,byrow=T)
B = matrix(c(1,-5,-3,4,-2,2), 3,2,byrow=T)
C = A*B
```

Definition 12. Kronecker product of two matrices

$$\mathbf{A}_{k \times m} \otimes \mathbf{B}_{n \times s} = \begin{bmatrix} a_{11} \mathbf{B} & a_{12} \mathbf{B} & \cdots & a_{1m} \mathbf{B} \\ a_{21} \mathbf{B} & a_{22} \mathbf{B} & \cdots & a_{2m} \mathbf{B} \\ \vdots & \vdots & & \vdots \\ a_{k1} \mathbf{B} & a_{k2} \mathbf{B} & \cdots & a_{km} \mathbf{B} \end{bmatrix}$$

Example 9.

$$\begin{bmatrix} 2 & 4 \\ 0 & -2 \\ 3 & -1 \end{bmatrix} \otimes \begin{bmatrix} 5 & 3 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 10 & 6 & 20 & 12 \\ 4 & 2 & 8 & 4 \\ 0 & 0 & -10 & -6 \\ 0 & 0 & -4 & -2 \\ 15 & 9 & -5 & -3 \\ 6 & 3 & -2 & -1 \end{bmatrix}$$

$$\mathbf{a} \otimes \mathbf{y} = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} \otimes \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} a_1 y_1 \\ a_1 y_2 \\ a_2 y_1 \\ a_2 y_2 \\ a_3 y_1 \\ a_3 y_2 \end{bmatrix}$$

R-codes:

```
A = matrix(c(2,4,0,-2,3,-1),ncol=2,byrow=T)
B = matrix(c(5,3,2,1),2,2,byrow=T)
C = kronecker(A,B)
```


1.3 Determinant

Definition 13. The **determinant** of an $n \times n$ matrix $\mathbf{A}$ is

$$|\mathbf{A}| = \sum_{j=1}^n a_{ij}(-1)^{i+j}|M_{ij}| \text{ for any row } i$$

or

$$|\mathbf{A}| = \sum_{i=1}^n a_{ij}(-1)^{i+j}|M_{ij}| \text{ for any column } j$$

where M_{ij} is the “minor” for a_{ij} obtained by deleting the i^{th} row and j^{th} column from A .

Example 10.

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

$$|\mathbf{A}| = a_{11}(-1)^{1+1}|a_{22}| + a_{12}(-1)^{1+2}|a_{21}|$$

$$\text{then } \begin{vmatrix} 7 & 2 \\ 4 & 5 \end{vmatrix} =$$

Example 11.

$$\begin{aligned} |\mathbf{A}| &= \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} \\ &= a_{11}(-1)^{1+1} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} + a_{12}(-1)^{1+2} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} \\ &\quad + a_{13}(-1)^{1+3} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix} \end{aligned}$$

$$\text{then } \begin{vmatrix} 1 & 1 & 3 \\ 4 & 3 & 6 \\ 7 & 5 & 9 \end{vmatrix} =$$

```
R-codes:
A = matrix(c(1,1,3,4,3,6,7,5c',9),3,3,byrow=T)
> detA = det(A)
> detA
```

Properties of determinants

- $|\mathbf{A}^T| = |\mathbf{A}|$
- $|\mathbf{A}|$ = product of the eigenvalues of $\mathbf{A}$
- $|\mathbf{AB}| = |\mathbf{A}||\mathbf{B}|$ when $\mathbf{A}$ and $\mathbf{B}$ are square matrices of the same order.
- $\begin{vmatrix} \mathbf{P} & \mathbf{0} \\ \mathbf{X} & \mathbf{Q} \end{vmatrix} = |\mathbf{P}||\mathbf{Q}|$ when $\mathbf{P}$ and $\mathbf{Q}$ are square matrices of the same order and $\mathbf{0}$ is a matrix of zeros.
- $|\mathbf{AB}| = |\mathbf{BA}|$ when the matrix product is defined
- $|c\mathbf{A}| = c^k|\mathbf{A}|$ when c is a scalar and $\mathbf{A}$ is a $k \times k$ matrix

1.4 Orthogonal and Idempotent Matrices

Definition 14. A square matrix $\mathbf{A}$ is said to be **orthogonal** if

$$\mathbf{A}\mathbf{A}^T = \mathbf{A}^T\mathbf{A} = \mathbf{I} \quad (\text{then } \mathbf{A}^{-1} = \mathbf{A}^T)$$

Definition 15. A square matrix $\mathbf{P}$ is **idempotent** if $\mathbf{PP} = \mathbf{P}$

Example 12. (Orthogonal Matrix)

$$\mathbf{A} = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} \quad \mathbf{A} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

In each case the columns of $\mathbf{A}$ are coefficients for orthogonal contrasts.

Example 13. (Idempotent Matrix)

$$\mathbf{P} = \begin{bmatrix} \frac{5}{6} & \frac{2}{6} & -\frac{1}{6} \\ \frac{2}{6} & \frac{2}{6} & \frac{2}{6} \\ -\frac{1}{6} & \frac{2}{6} & \frac{5}{6} \end{bmatrix}$$

1.5 Linear Combinations and Column Spaces

$\mathbf{A}\mathbf{b}$ is a linear combination of the columns of an $m \times n$ matrix $\mathbf{A}$.

$$\mathbf{A}\mathbf{b} = [\mathbf{a}_1, \dots, \mathbf{a}_n] \begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix} = b_1\mathbf{a}_1 + \dots + b_n\mathbf{a}_n$$

The set of all possible linear combinations of the columns of $\mathbf{A}$ is called the column space of $\mathbf{A}$ and is written as

$$\mathcal{C}(\mathbf{A}) = \{\mathbf{A}\mathbf{b} : \mathbf{b} \in \mathbf{R}^n\}$$

Note that $\mathcal{C}(\mathbf{A}) \subseteq \mathbf{R}^m$.

Example 14.

Suppose $\mathbf{X}$ is an $n \times p$ matrix and $\mathbf{B}$ is a $p \times p$ non-singular matrix. Prove that $\mathcal{C}(\mathbf{X}) = \mathcal{C}(\mathbf{X}\mathbf{B}^{-1})$.

1.6 Linear Independence

Definition 16. A set of n -dimensional vectors $\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_k$ are **linearly independent** if there is no set of scalars $a_1 \ a_2 \ \dots \ a_k$ such that

$$\mathbf{0} = \sum_{j=1}^k a_j \mathbf{y}_j$$

and at least one a_j is non-zero.

Example 15. Show that

$$\mathbf{y}_1 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \quad \mathbf{y}_2 = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} \quad \mathbf{y}_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

are linearly independent.

Example 16. Show that

$$\mathbf{y}_1 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \quad \mathbf{y}_2 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \quad \mathbf{y}_3 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

are not linearly independent.

1.7 Rank

Definition 17. The **row rank** of a matrix is the number of linearly independent rows, where each row is considered as a vector.

Definition 18. The **column rank** of a matrix is the number of linearly independent columns, with each column considered as a vector.

Example 17. Show that the row and column rank of

$$\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 5 & -1 \\ 0 & 1 & -1 \end{bmatrix}$$

is 2.

R-codes:

```
A = matrix(c(1,1, 1,2,5,-1,0,1,-1),3,3,byrow=T)
rA = qr(A)$rank
```

Result 1. The row rank and the column rank of a matrix are equal.

Definition 19. The **rank** of a matrix is either the row rank or the column rank of the matrix.

Definition 20. A square matrix $\mathbf{A}_{k \times k}$ is **non-singular** if its rank is equal to the number of rows (or columns).

This is equivalent to the condition

$$\mathbf{A}_{k \times k} \mathbf{b}_{k \times 1} = \mathbf{0}_{k \times 1} \text{ only when } \mathbf{b} = \mathbf{0}$$

A matrix that fails to be nonsingular is called **singular**.

Result 2. If $\mathbf{B}_{n \times n}$ is non-singular and $\mathbf{A}_{n \times m}$, then

$$\text{rank}(\mathbf{BA}) = \text{rank}(\mathbf{A}).$$

Result 3. If $\mathbf{B}$ and $\mathbf{C}$ are non-singular matrices and products with $\mathbf{A}$ are defined, then

$$\text{rank}(\mathbf{BA}) = \text{rank}(\mathbf{AC}) = \text{rank}(\mathbf{A}).$$

Result 4. $\text{rank}(\mathbf{A}^T \mathbf{A}) = \text{rank}(\mathbf{A}) = \text{rank}(\mathbf{A}^T)$.

1.8 Inverse

Definition 21. The **identity matrix**, denoted by $\mathbf{I}$, is a $k \times k$ matrix of the form

$$\mathbf{I} = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 1 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & 1 \end{bmatrix}$$

Definition 22. The **inverse** of a square, non-singular matrix $\mathbf{A}$ is the matrix, denoted by $\mathbf{A}^{-1}$, such that

$$\mathbf{AA}^{-1} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{I}$$

```
R-codes:
I3 = diag(rep(1,3))
I3
W = matrix(c(1,2,3,4,5,6,7,8,10),3,3,byrow=T)
Winv = solve(W)
Winv
```

Result 5.

(i) The inverse of $\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ is

$$\mathbf{A}^{-1} = \frac{1}{|\mathbf{A}|} \begin{bmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{bmatrix}$$

(ii) In general, the (i, j) element of $\mathbf{A}^{-1}$ is

$$\frac{(-1)^{i+j} |\mathbf{A}_{ji}|}{|\mathbf{A}|}$$

where $\mathbf{A}_{ji}$ is the matrix obtained by deleting the j -th row and i -th column of $\mathbf{A}$.

Result 6. For a $k \times k$ matrix $\mathbf{A}$, the following are equivalent:

- (i) $\mathbf{A}$ is nonsingular
- (ii) $|\mathbf{A}| \neq 0$
- (iii) $\mathbf{A}^{-1}$ exists

Result 7. For $k \times k$ nonsingular matrices $\mathbf{A}$ and $\mathbf{B}$

- (i) $(\mathbf{A}^T)^{-1} = (\mathbf{A}^{-1})^T$
- (ii) $(\mathbf{A}\mathbf{B})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$
- (iii) $|\mathbf{A}^{-1}| = 1/|\mathbf{A}|$
- (iv) $\mathbf{A}^{-1}$ is unique and nonsingular
- (v) $(\mathbf{A}^{-1})^{-1} = \mathbf{A}$
- (vi) If $\mathbf{A}$ is symmetric, then $\mathbf{A}^{-1}$ is symmetric

Result 8. Inverse of a Diagonal Matrix

$$\begin{bmatrix} a_{11} & & & \\ & a_{22} & & \\ & & \ddots & \\ & & & a_{kk} \end{bmatrix}^{-1} = \begin{bmatrix} 1/a_{11} & & & \\ & 1/a_{22} & & \\ & & \ddots & \\ & & & 1/a_{kk} \end{bmatrix}$$

Result 9.

If $\mathbf{B}$ is a $k \times k$ non-singular matrix and $\mathbf{B} + \mathbf{c}\mathbf{c}^T$ is non-singular, then

$$(\mathbf{B} + \mathbf{c}\mathbf{c}^T)^{-1} = \mathbf{B}^{-1} - \frac{\mathbf{B}^{-1}\mathbf{c}\mathbf{c}^T\mathbf{B}^{-1}}{1 + \mathbf{c}^T\mathbf{B}^{-1}\mathbf{c}}$$

Result 10.

Let $\mathbf{I}_n$ be an $n \times n$ identity matrix and let $\mathbf{J}_n = \mathbf{1}\mathbf{1}^T$ be an $n \times n$ matrix where each element is one, then

$$(a\mathbf{I}_n + b\mathbf{J}_n)^{-1} = \frac{1}{a} \left(\mathbf{I}_n - \frac{b}{a + nb} \mathbf{J}_n \right)$$

Example 18.

Suppose $\mathbf{Z} = \mathbf{1}_{6 \times 1}$, $\mathbf{G} = 25$, $\mathbf{R} = 49 \mathbf{I}_{6 \times 6}$. If $\mathbf{\Sigma} = \mathbf{ZGZ}^T + \mathbf{R}$, find $\mathbf{\Sigma}^{-1}$.

1.9 Trace

Definition 23. The **trace** of a $k \times k$ matrix $\mathbf{A} = \{a_{ij}\}$ is the sum of the diagonal elements:

$$tr(\mathbf{A}) = \sum_{j=1}^k a_{jj}$$

R-codes:

```
W = matrix(c(1,2,3,4,5,6,7,8,10),
            3,3,byrow=T)
trW = sum(diag(W))
trW
```

R-codes:

```
W = {1 2 3, 4 5 6, 7 8 10};
trW1 = trace(W);
trW2 = sum(diag(W));
print W trW1 trW2;
```

Result 11. Let $\mathbf{A}$ and $\mathbf{B}$ denote $k \times k$ matrices and let c be a scalar. Then,

- (i) $tr(c \mathbf{A}) = c tr(\mathbf{A})$
- (ii) $tr(\mathbf{A} + \mathbf{B}) = tr(\mathbf{A}) + tr(\mathbf{B})$
- (iii) $tr(\mathbf{AB}) = tr(\mathbf{BA})$
- (iv) $tr(\mathbf{B}^{-1}\mathbf{AB}) = tr(\mathbf{A})$
- (v) $tr(\mathbf{A} \mathbf{A}^T) = \sum_{i=1}^k \sum_{j=1}^k a_{ij}^2$

Example 19.

For $\mathbf{A} = \mathbf{I}_{n \times n} - \frac{1}{n} \mathbf{i} \mathbf{i}^T$ where $\mathbf{i}^T = [1 \ 1 \ \cdots \ 1]_{1 \times n}$.

- (a) Show that $\mathbf{A}$ is idempotent.
- (b) Find $tr(\mathbf{A})$.
- (c) Interpret the result of $\mathbf{A} \mathbf{y}$ where $\mathbf{y}$ is $n \times 1$.

1.10 Eigenvalues and Eigenvectors

Definition 24. For a $k \times k$ matrix $\mathbf{A}$, the scalars $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_k$ satisfying the polynomial equation

$$|\mathbf{A} - \lambda \mathbf{I}| = 0$$

are called the eigenvalues (or characteristic roots) of $\mathbf{A}$.

Definition 25. Corresponding to any eigenvalue λ_i is an eigenvector (or characteristic vector) $\mathbf{u}_i \neq \mathbf{0}$ satisfying

$$\mathbf{A} \mathbf{u}_i = \lambda_i \mathbf{u}_i.$$

Comment: Eigenvectors are not unique

- (i) If $\mathbf{u}_i$ is an eigenvector for λ_i , then $c \mathbf{u}_i$ is also an eigenvector for any scalar $c \neq 0$.

(ii) We will adopt the following conventions (for real symmetric matrices)

$$\mathbf{u}_i^T \mathbf{u}_i = 1 \quad \text{for all } i = 1, \dots, k$$

$$\mathbf{u}_i^T \mathbf{u}_j = 0 \quad \text{for all } i \neq j$$

(iii) Even with (ii), eigenvectors are not unique

- If $\mathbf{u}_i$ is an eigenvector satisfying (ii), then $-\mathbf{u}_i$ is also an eigenvector satisfying (ii).
- If $\lambda_i = \lambda_j$ then there are an infinite number of choices for $\mathbf{u}_i$ and $\mathbf{u}_j$.

Result 12. For a $k \times k$ symmetric matrix $\mathbf{A}$ with elements that are real numbers

- (i) every eigenvalue of $\mathbf{A}$ is a real number
- (ii) $\text{rank}(\mathbf{A}) = \text{number of non-zero eigenvalues}$
- (iii) if $\mathbf{A}$ is non-negative definite, then $\lambda_i \geq 0$ for all $i = 1, 2, \dots, k$
- (iv) if $\mathbf{A}$ is positive definite then $\lambda_i > 0$ for all $i = 1, 2, \dots, k$

$$(v) \text{trace}(\mathbf{A}) = \sum_{i=1}^k a_{ii} = \sum_{i=1}^k \lambda_i$$

$$(vi) |\mathbf{A}| = \prod_{i=1}^k \lambda_i$$

- (vii) if $\mathbf{A}$ is idempotent ($\mathbf{A} \mathbf{A} = \mathbf{A}$), then the eigenvalues are either zero or one.

1.11 Quadratic Form

Definition 26.

Let $\mathbf{A}$ be a $k \times k$ matrix and let $\mathbf{y}$ be a vector of order k , then

$$\mathbf{y}^T \mathbf{A} \mathbf{y} = \sum_{i=1}^k \sum_{j=1}^k y_i y_j a_{ij}$$

is called a **quadratic form**.

Suppose $\mathbf{y}_{n \times 1}$ is a vector of n observations. Then $\mathbf{y}^T \mathbf{y} = \sum_{i=1}^n y_i^2$ is the total sum of squares of the observations. Let $\mathbf{P}$ be an orthogonal matrix

$$\mathbf{P} \mathbf{P}^T = \mathbf{P}^T \mathbf{P} = \mathbf{I}$$

and partition $\mathbf{P}$ row wise into k sub-matrices $\mathbf{P}_i$, of order $n_i \times n$, for $i = 1, 2, \dots, k$, with $\sum_{i=1}^k n_i = n$; i.e.

$$\mathbf{P} = \begin{bmatrix} \mathbf{P}_1 \\ \mathbf{P}_2 \\ \vdots \\ \mathbf{P}_k \end{bmatrix} \quad \text{and} \quad \mathbf{P}^T = [\mathbf{P}_1^T \ \mathbf{P}_2^T \ \dots \ \mathbf{P}_k^T].$$

Then

$$\mathbf{y}^T \mathbf{y} = \mathbf{y}^T \mathbf{I} \mathbf{y} = \mathbf{y}^T \mathbf{P}^T \mathbf{P} \mathbf{y} = \sum_{i=1}^k \mathbf{y}^T \mathbf{P}_i^T \mathbf{P}_i \mathbf{y}.$$

In this way $\mathbf{y}^T \mathbf{y}$ is partition into k sums of squares

$$\mathbf{y}^T \mathbf{P}_i^T \mathbf{P}_i \mathbf{y} \quad \text{for } i = 1, \dots, k$$

each of these sums of squares corresponds to the lines in an analysis of variance, having $\mathbf{y}^T \mathbf{y}$ as the total sums of squares.

Example 20.

Corresponding to a vector of 4 observations consider

$$\mathbf{P} = \begin{bmatrix} \frac{1}{\sqrt{4}} & \frac{1}{\sqrt{4}} & \frac{1}{\sqrt{4}} & \frac{1}{\sqrt{4}} \\ \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} & 0 & 0 \\ \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{6}} & \frac{-2}{\sqrt{6}} & 0 \\ \frac{1}{\sqrt{12}} & \frac{1}{\sqrt{12}} & \frac{1}{\sqrt{12}} & \frac{-3}{\sqrt{12}} \end{bmatrix} = \begin{bmatrix} \mathbf{P}_1 \\ \mathbf{P}_2 \end{bmatrix}$$

Show that $\mathbf{P}$ is orthogonal and find the two partition sums of squares.

1.11.1 Helmert Matrix

The matrix $\mathbf{P}$ in Example 20 is a fourth-order Helmert matrix. Writing

$$\mathbf{H}_{n \times n} = \begin{bmatrix} \mathbf{h}^T \\ \mathbf{H}_0 \end{bmatrix} \begin{matrix} 1 \times n \\ (n-1) \times n \end{matrix}$$

for a Helmert matrix of order n ,

$$\mathbf{h}^T = \text{first row of } \mathbf{H}_{n \times n} = \frac{1}{\sqrt{n}} \mathbf{1}_n^T$$

and

$$\mathbf{H}_0 = \text{last } (n-1) \text{ rows of } \mathbf{H}_{n \times n}$$

with $\mathbf{H}_0$ having its r th row as

$$\begin{bmatrix} \frac{1}{\sqrt{r(r+1)}} \mathbf{1}_r^T & \frac{-r}{\sqrt{r(r+1)}} \mathbf{0}_{(n-r-1) \times 1} \end{bmatrix}$$

$$r = 1, 2, \dots, n-1$$

1. Is $\mathbf{H}_{n \times n}$ orthogonal?
2. What is $\mathbf{y}^T \mathbf{h} \mathbf{h}^T \mathbf{y}$ represent?
3. What is $\mathbf{y}^T \mathbf{H}_0^T \mathbf{H}_0 \mathbf{y}$ represent?

1.11.2 Symmetric Matrices

Any quadratic form $\mathbf{y}^T \mathbf{A} \mathbf{y}$ can be written as $\mathbf{y}^T \mathbf{A} \mathbf{y} = \mathbf{y}^T \mathbf{B} \mathbf{y}$ where $\mathbf{B} = \frac{1}{2}(\mathbf{A} + \mathbf{A}^T)$ is symmetric. Furthermore, any quadratic form can be written as $\mathbf{y}^T \mathbf{A} \mathbf{y}$ for an infinite number of matrices, but can only be written in one way as $\mathbf{y}^T \mathbf{B} \mathbf{y}$ for $\mathbf{B}$ symmetric. For example,

$$4y_1^2 + 6y_1y_2 + 7y_2^2 = \begin{bmatrix} y_1 & y_2 \end{bmatrix} \begin{bmatrix} 4 & 3+a \\ 3-a & 7 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

for any value of a , but only when $a = 0$ is the matrix involved symmetric. This means that for any particular quadratic form there is only one, unique matrix such that the quadratic form can be written as $\mathbf{y}^T \mathbf{A} \mathbf{y}$ with $\mathbf{A}$ being symmetric. Due to the uniqueness of this symmetric matrix, the quadratic form that we are going to discuss is confined to the case of $\mathbf{A}$ being symmetric.

1.11.3 Positive Definiteness

Definition 27.

A quadratic form $\mathbf{y}^T \mathbf{A} \mathbf{y}$ is said to be **positive definite** (p.d.) if

$$\mathbf{y}^T \mathbf{A} \mathbf{y} > 0 \text{ for all } \mathbf{y} \text{ except } \mathbf{y} = \mathbf{0}.$$

The corresponding (symmetric) matrix is also described as positive definite.

Definition 28. A quadratic form $\mathbf{y}^T \mathbf{A} \mathbf{y}$ is said to be **positive semi-definite** (p.s.d) if

$$\mathbf{y}^T \mathbf{A} \mathbf{y} \geq 0 \text{ for all } \mathbf{y} \neq \mathbf{0}$$

with $\mathbf{y}^T \mathbf{A} \mathbf{y} = 0$ for at least one $\mathbf{y} \neq \mathbf{0}$.

The corresponding (symmetric) matrix $\mathbf{A}$ is a p.s.d. matrix.

Example 21. Show that

$$\mathbf{A} = \begin{pmatrix} 3 & 5 & 1 \\ 5 & 13 & 0 \\ 1 & 0 & 1 \end{pmatrix}$$

is a positive definite matrix.

Example 22.

Show that

$$\mathbf{B} = \begin{pmatrix} 13 & -2 & -3 \\ -2 & 10 & -6 \\ -3 & -6 & 5 \end{pmatrix}$$

is a positive semidefinite matrix.

1.12 Spectral Decomposition

Result 13. The spectral decomposition of a $k \times k$ symmetric matrix $\mathbf{A}$ with eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_k$ and eigenvectors $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k$ (with $\mathbf{u}_i^T \mathbf{u}_i = 1$ and $\mathbf{u}_i^T \mathbf{u}_j = 0$) is

$$\begin{aligned} \mathbf{A} &= \lambda_1 \mathbf{u}_1 \mathbf{u}_1^T + \lambda_2 \mathbf{u}_2 \mathbf{u}_2^T + \dots + \lambda_k \mathbf{u}_k \mathbf{u}_k^T \\ &= \mathbf{U} \mathbf{D} \mathbf{U}^T \end{aligned}$$

where

$$\mathbf{D} = \begin{bmatrix} \lambda_1 & & \\ & \lambda_2 & \\ & & \ddots \\ & & & \lambda_k \end{bmatrix}$$

and

$$\mathbf{U} = [\mathbf{u}_1 \mid \mathbf{u}_2 \mid \dots \mid \mathbf{u}_k]$$

is an orthogonal matrix.

Result 14. If $\mathbf{A}$ is a $k \times k$ symmetric nonsingular matrix with spectral decomposition

$$\mathbf{A} = \sum_{i=1}^k \lambda_i \mathbf{u}_i \mathbf{u}_i^T = \mathbf{U} \mathbf{D} \mathbf{U}^T$$

then

$$(i) \mathbf{A}^{-1} = \sum_{i=1}^k \lambda_i^{-1} \mathbf{u}_i \mathbf{u}_i^T = \mathbf{U} \mathbf{D}^{-1} \mathbf{U}^T$$

(ii) the square root matrix

$$\mathbf{A}^{1/2} = \sum_{i=1}^k \sqrt{\lambda_i} \mathbf{u}_i \mathbf{u}_i^T$$

has the properties:

$$(a) \mathbf{A}^{1/2} \mathbf{A}^{1/2} = \mathbf{A}$$

$$(b) \mathbf{A}^{1/2} \mathbf{A}^{-1} \mathbf{A}^{1/2} = \mathbf{I}$$

$$(c) \mathbf{A}^{1/2} \text{ is symmetric}$$

(iii) The inverse square root matrix

$$\begin{aligned} \mathbf{A}^{-1/2} &= \sum_{i=1}^k \frac{1}{\sqrt{\lambda_i}} \mathbf{u}_i \mathbf{u}_i^T \\ &= \mathbf{U} \mathbf{D}^{-1/2} \mathbf{U}^T \end{aligned}$$

has the properties:

$$(a) \mathbf{A}^{-1/2} \mathbf{A}^{-1/2} = \mathbf{A}^{-1}$$

$$(b) \mathbf{A}^{-1/2} \mathbf{A} \mathbf{A}^{-1/2} = \mathbf{I}$$

$$(c) \mathbf{A}^{-1/2} \text{ is symmetric}$$

In parts (ii) and (iii), $\mathbf{A}$ should be positive definite to ensure that

$$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_k > 0$$

1.13 Random Vectors:

Definition 29.

A random vector $\mathbf{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}$ is a vector whose elements are random variables.

1.13.1 Mean vectors:

$$E(\mathbf{y}) = \begin{bmatrix} E(y_1) \\ \vdots \\ E(y_n) \end{bmatrix} = \begin{bmatrix} \mu_1 \\ \vdots \\ \mu_n \end{bmatrix} = \boldsymbol{\mu}$$

where

$$\mu_i = E(y_i) = \int_{-\infty}^{\infty} y f_i(y) dy$$

if y_i is a continuous random variable with density function $f_i(y)$

and

$$\mu_i = E(y_i) = \sum y p_i(y)$$

if y_i is a discrete random variable with probability function $p_i(y)$.

1.13.2 Covariance matrix:

$$\boldsymbol{\Sigma} = Var(\mathbf{y}) = \begin{bmatrix} \sigma_1^2 & \sigma_{12} & \sigma_{13} & \cdots & \sigma_{1n} \\ \sigma_{21} & \sigma_2^2 & \sigma_{23} & \cdots & \sigma_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \sigma_{n1} & \sigma_{n2} & \sigma_{n3} & \cdots & \sigma_n^2 \end{bmatrix}$$

with variances

$$\begin{aligned} Var(y_i) &= \sigma_i^2 = E(y_i - \mu_i)^2 \\ &= \begin{cases} \int_{-\infty}^{\infty} (y - \mu_i)^2 f_i(y) dy & \text{if } y \text{ is a continuous} \\ & \text{random variable} \\ \sum_{all\ y} (y - \mu_i)^2 p_i(y) & \text{if } y \text{ is a discrete} \\ & \text{random variable} \end{cases} \end{aligned}$$

and covariances:

$$\sigma_{ij} = Cov(y_i, y_j) = E[(y_i - \mu_i)(y_j - \mu_j)]$$

where

$$\sigma_{ij} = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (y - \mu_i)(v - \mu_j) f_{ij}(y, v) dy dv$$

if y_i and y_j are continuous random variables with joint density function $f_{ij}(y, v)$ and

$$\sigma_{ij} = \sum_{\substack{\text{all} \\ y}} \sum_{\substack{\text{all} \\ v}} (y - \mu_i)(v - \mu_j) P_{ij}(y, v)$$

if y_i and y_j are discrete random variables with joint probability function

$$p_{ij}(y, v) = Pr(y_i = y, V_j = v)$$

Result 15.

Let $\mathbf{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}$ be a random vector with

$$\boldsymbol{\mu} = E(\mathbf{y}) \quad \text{and} \quad \boldsymbol{\Sigma} = Var(\mathbf{y}),$$

and let

$$\mathbf{A}_{p \times n} = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{p1} & \cdots & a_{pn} \end{bmatrix}$$

be a matrix of non-random elements, and let

$$\mathbf{c} = \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} \quad \text{and} \quad \mathbf{d} = \begin{bmatrix} d_1 \\ \vdots \\ d_p \end{bmatrix}$$

be vectors of non-random elements, then

- (i) $E(\mathbf{A}\mathbf{y} + \mathbf{d}) = \mathbf{A}\boldsymbol{\mu} + \mathbf{d}$
- (ii) $Var(\mathbf{A}\mathbf{y} + \mathbf{d}) = \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^T$
- (iii) $E(\mathbf{c}^T \mathbf{y}) = \mathbf{c}^T \boldsymbol{\mu}$
- (iv) $Var(\mathbf{c}^T \mathbf{y}) = \mathbf{c}^T \boldsymbol{\Sigma} \mathbf{c}$

Example 23.

Let the 3×1 random vector $\mathbf{y}$ follows a multi-variate normal distribution with mean vector $\boldsymbol{\mu} = [7 \ 9 \ 5]^T$ and covariance matrix Σ where

$$\Sigma = \sigma^2 \begin{bmatrix} 1 & 0 & -1 \\ 0 & 4 & 0 \\ -1 & 0 & 3 \end{bmatrix}$$

Consider the vector $\mathbf{w}$ where

$$\mathbf{w} = \begin{bmatrix} 3y_1 - y_2 + 2y_3 - 25 \\ 2y_1 + y_2 - 4y_3 - 12 \end{bmatrix}$$

- (a) Find the mean vector of $\mathbf{w}$.
- (b) Find the covariance matrix of $\mathbf{w}$.